

PRD: ACM Student Chapter Scroll-Journey Website

Audience for this document: an AI coding agent that will build or extend the website. It is written to be self-contained: all copy is included in the Content Spec, so no other file is needed.

1. Goal

Build a single-page, scroll-driven website for the ACM student chapter. As the visitor scrolls, a boy travels from home to college (walk, bus, metro, boat, college door). Each stop of the journey reveals one section of the chapter's content. The finished site must be fast, readable on phones, and easy for the chapter team to edit.

Success means: all copy in section 6 appears on the site, word for word; the journey animation runs smoothly; and every section is reachable from the navigation.

2. Scope

In scope: the single scroll page, navigation, all content sections below, the Join and Notify Me forms (front end only), a Team section at the end, responsive layout, accessibility basics, a content file the team can edit.

Out of scope for v1: user accounts, a database, an admin panel, payments, a blog, dynamic event loading from an API.

3. Recommended stack (defaults, change only if the project already uses something else)

- Vite + React (or plain HTML/CSS/JS if the existing site is static)
- GSAP + ScrollTrigger for the journey; CSS for everything else
- Illustrations as inline SVG or optimized WebP/PNG; no video
- Tailwind or plain CSS with design tokens (see section 5)
- Form handling through a placeholder endpoint or a free form service (Formspree, Google Forms embed); keep it swappable

4. Journey map (proposed, adjustable)

Stop	Scene	Section shown	Content source
0	Home, boy leaves the door	Hero	6.1
1	Catches the bus	About the chapter + What is ACM	6.2, 6.3
2	Walks to the station, rides the metro	Roadmap + Upcoming Activities (events)	6.4, 6.5
3	Crosses by boat	Focus Areas	6.6
4	Walks up to the college	Vision and Mission	6.7
5	Opens the college door	Join CTA, then Team (members appear one by one)	6.8, 6.9

The scene changes (bus, metro, boat) are the transitions between sections. Content cards slide in over or beside the moving scene. The boy character moves along a path as scroll progress changes.

5. Global requirements

Navigation: fixed top bar with links About, Events, Focus Areas, Vision, Mission, Team, Join. Clicking a link scrolls to that stop and plays the journey up to it. Show a thin scroll-progress bar.

Design tokens (define as CSS variables, tune after seeing the reference photo): one display font and one body font; a background, a text color, one accent, one muted color; spacing scale 8, 16, 24, 48, 96 px; light and dark values.

Responsive: design for 390 px wide first, then tablet and desktop. On small screens simplify the journey (smaller scene, fewer layers) but keep the order of stops.

Performance: target Lighthouse performance 85+ on mobile; lazy-load images below the first scene; animate only transform and opacity; keep total JS under about 300 KB gzipped.

Accessibility: honor prefers-reduced-motion (no scroll-scrubbed animation, stack sections normally); all text at least WCAG AA contrast; real headings in order; alt text on every meaningful image; keyboard focus visible; forms with labels.

SEO: page title "ACM Student Chapter"; meta description from the About paragraph; semantic sections with ids (hero, about, events, focus, vision, mission, team, join).

6. Content Spec (use word for word)

6.1 Hero (id: hero)

- Eyebrow: Official Student Chapter
- H1: ACM NMIET Student Chapter
- Subtitle: Innovating for the Future | Est. 2025
- Buttons: Explore More (goes to the About stop), View Events (goes to the Events stop)
- Nav labels: About, Vision, Mission, Roadmap

6.2 About (id: about)

Heading: Welcome to ACM NMIET

Established in 2025, the ACM NMIET Student Chapter is dedicated to fostering technical growth and student excellence through innovation and collaboration.

We provide a platform for students to explore the latest in computing, connect with industry experts, and build projects that matter. Our community thrives on the curiosity of our members and the drive to solve real-world problems.

Stats (animated counters): 2025 Established; 15+ Active Members; 20+ Events Yearly.

6.3 What is ACM (id: what-is-acm)

Heading: What is ACM? Subheading: Understanding the world's largest educational and scientific computing society.

Overview, Association for Computing Machinery:

The Association for Computing Machinery (ACM) is the world's largest educational and scientific computing society, uniting educators, researchers, and professionals to inspire dialogue, share resources, and address the field's challenges. ACM strengthens the computing profession's collective voice through strong leadership, promotion of the highest standards, and recognition of technical excellence.

Founded at the dawn of the computer age, ACM's reach extends to every part of the globe, with more than half of its 100,000 members residing outside the United States. Its growing membership has led to Councils in Europe, India, and China, fostering networking opportunities that strengthen ties within and across countries and technical communities.

ACM provides the computing field's premier Digital Library and serves its members and the computing profession with leading-edge publications, conferences, and career resources.

Stats: 1947 Founded; 100k+ Members; Global Chapters.

Why ACM Matters for Students. Intro: Joining an ACM student chapter opens doors to a world of opportunities in computing.

- **Networking:** Connect with peers, mentors, and industry professionals. Build relationships that can last a lifetime and help your career.
- **Career Growth:** Access exclusive job boards, career advice, and professional development resources tailored for computing students.
- **Research Access:** Gain full access to the ACM Digital Library, the world's most comprehensive database of computing literature.
- **Mentorship:** Learn from distinguished speakers and participate in mentorship programs that guide your academic journey.
- **Events & Competitions:** Participate in hackathons, coding contests like ICPC, and technical workshops organized by local chapters.
- **Skill Building:** Develop leadership, management, and technical skills by taking on roles within the student chapter committee.

About ACM Student Chapters:

ACM Student Chapters are the local hubs of activity for ACM members on university campuses worldwide. They serve as a gateway to the larger professional network and provide a platform for students to organize technical talks, workshops, hackathons, and social events.

By joining the NMIET Student Chapter, you become part of this global family, gaining resources to enhance your learning beyond the classroom and preparing you for a successful career in technology.

Explore ACM Resources (external links, open in a new tab): ACM.org, Official Website (<https://www.acm.org>); Digital Library, Research Database (<https://dl.acm.org>); Learning Center, Courses & Webinars (<https://learning.acm.org>); Career Center, Job Board

(<https://jobs.acm.org>).

6.4 Roadmap (id: roadmap, shown during the metro ride)

Heading: Our Roadmap for Growth. Subheading: Key initiatives planned for the upcoming academic year.

- Technical Workshops: React & Tailwind Bootcamp (Q1 Foundation Series); AI/ML Intro Workshop (Q2 Specialized Track)
- Guest Lectures: Future of Cloud Computing (Speaker from AWS)
- Competitions: CodeWars 2025 (Inter-college Coding Contest); Design-a-thon (UI/UX Sprint)
- Community: Open Source Meetup (Hacktoberfest Prep)

6.5 Upcoming Activities (id: events, metro ride, after the roadmap)

Heading: Upcoming Activities. Subheading: Exciting initiatives we're planning for the semester. Stay ahead of the curve with our technical roadmap.

Note from the team. What's in the pipeline? We are currently curating a series of technical workshops and hackathons. Sign up to get notified as soon as dates are confirmed.

Form: one email field, button Notify Me. Success message: You're on the list! We'll notify you about upcoming events. Helper text: No spam, just updates on upcoming tech events.

Planned Activities: filter tabs All, Workshops, Competitions, Guest Lectures. The event list is empty in v1; render a friendly empty state and read events from a simple events array in the content file so the team can add them later.

6.6 Focus Areas (id: focus, shown during the boat ride)

Heading: Our Focus Areas. Subheading: Explore the diverse technical domains where we learn, build, and grow together. We focus on practical skills that bridge the gap between academia and industry.

Section heading: Technical Domains We're Passionate About. Intro: At ACM NMIET, we don't just talk about technology; we dive deep. From building scalable applications to securing digital infrastructures, our focus areas cover the most vital skills in today's tech landscape.

Six cards, each with a status badge and a technology tag list:

- Web & App Dev, Active. Building responsive websites and cross-platform mobile applications using modern frameworks and best practices. Technologies: React, Node.js, Flutter
- AI & Machine Learning, Active. Exploring the frontiers of intelligence, training data models, and implementing predictive algorithms to solve real problems. Technologies: Python, TensorFlow, PyTorch
- Competitive Coding, Active. Mastering algorithms and data structures to efficiently solve complex coding challenges and excel in competitions. Technologies: C++, Java, DSA
- Cybersecurity, Coming Soon. Learning to protect systems, networks, and programs from digital attacks through ethical hacking and cryptography. Technologies: Ethical Hacking, Linux
- UI/UX & Design, Active. Designing intuitive, accessible, and aesthetically pleasing user interfaces and seamless user experiences. Technologies: Figma, Prototyping
- Data Science, Planned. Unlocking actionable insights from raw data to drive decision-making and business innovation through analytics. Technologies: Pandas, SQL, NumPy

Closing call to action: Ready to learn with us? Join the ACM NMIET Chapter today and start your journey in one of these domains. Button: Join ACM Chapter.

6.7 Vision and Mission (ids: vision, mission, college approach)

Our Vision: "To be a leading center of innovation where technology meets creativity, empowering students to shape the digital future."

- Global Recognition: Connecting our members with the worldwide ACM network for opportunities and exposure.
- Student Excellence: Cultivating a standard of high academic and practical performance across all disciplines.
- Industry Integration: Bridging the gap between academic curriculum and real-world industrial requirements.

Our Mission: Empowering students through technology, mentorship, and practical experience.

We strive to create an inclusive environment that encourages learning, sharing, and building together. Button: Join Our Mission.

- Skill Development: Regular workshops on emerging tech stacks.
- Research: Encouraging participation in research papers and conferences.

- Peer Learning: Seniors mentoring juniors in a collaborative ecosystem.
- Networking: Connecting with alumni and professionals.

6.8 Join (id: join, at the college door)

The source site links its Join button to a separate Join page whose text was not captured. For v1, the door opens to a short form: name, email, year and branch, a Join button, and a success message. Mark the exact copy as TODO for the team to supply.

6.9 Team (id: team, after the door opens)

After the door opens, member cards appear one by one as the user keeps scrolling. Team names, roles and photos were not provided: build the component from a members array (name, role, photo, optional links) and ship with 4 clearly marked placeholder members.

7. Content file (so the team can edit without touching components)

Create one file, `content.json` (or `content.ts`), with these top-level keys: `site` (`chapterName`, `tagline`, `established`), `hero`, `about`, `whatIsAcm`, `roadmap`, `events`, `focusAreas`, `vision`, `mission`, `join`, `team`, `links`. Components must read all text from this file; no copy hard-coded in components.

8. Animation spec

- One master GSAP timeline scrubbed by `ScrollTrigger`, with a labeled position per stop (home, bus, metro, boat, college, door).
- The boy moves along an SVG path; at each transport stop the scene background swaps with a cross-fade or slide, and a vehicle element (bus, metro, boat) moves across.
- Content cards enter with opacity and a short upward translate; each plays once, not on every scroll.
- Pin the scene during a stop only as long as its content needs (about 100 to 200 vh).
- Refresh `ScrollTrigger` on resize; kill timelines on unmount.

9. Assets needed from the team

Reference photo for the theme; chapter logo; brand colors if any; character and scene illustrations (or approval to draw simple SVG ones); real team list with photos; event details when confirmed; the Join page copy.

10. Acceptance criteria

1. Every string in section 6 appears on the page exactly as written.
2. Navigation links scroll to the right stop; the progress bar tracks scroll.
3. Stats counters animate once when in view.
4. Focus Areas cards show the correct status badge for each domain.
5. Notify Me and Join forms validate email and show the success messages.
6. Works at 390 px, 768 px and 1440 px with no horizontal scroll.
7. With reduced motion on, the page is a plain stacked layout with all content readable.
8. Lighthouse mobile: performance 85+, accessibility 95+.

11. Build order for the AI

1. Scaffold the project and content.json with all copy from section 6.
2. Build static sections in order with the design tokens, no animation.
3. Add navigation, progress bar and the forms.
4. Add the scene layers and the master scroll timeline.
5. Add the Team reveal and the door transition.
6. Optimize, test the acceptance criteria, fix.

After each step, run the site and confirm it before moving on.

12. Open questions for the team

1. Chapter name on the site: the copy above says NMJET; change the chapterName in content.json if the site is for another college.
2. Is the journey order in section 4 correct, or should Vision and Mission come earlier?
3. What should the Join form do with submissions: email, Google Sheet, or a database later?